The Dedekind reals are Dedekind complete

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'RD-dedekind-complete' (source lines 648-655, Corollary 11.2.16); the proof environment starts after one blank line, so the pairing is by adjacency.

Editorial note. Two context spans are retained, namely the two results the proof invokes by label; both belong to the same section as the counted corollary, so they print with their own original numbers.

Editorial note. The retained counter values (chapter 11, section 2, theorem offset 13) make the first retained context print as Theorem 11.2.14, the second as Lemma 11.2.15 and the counted corollary as Corollary 11.2.16, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: Dedekind completeness of the Dedekind reals (Corollary 11.2.16).

Allowed standard prerequisites (editorial).

- Dedekind cut. A Dedekind cut (L,U) of rationals is a pair of predicates with the usual roundedness and disjointness conditions; RD is the type of such cuts.. It is the object of the statement and of the completion discussed in the proof.
- Dedekind completion and admissibility. The Dedekind completion of an ordered field is admissible for Omega when its cuts are again representable.. It is the content of the retained context lemma used at source line 658.
- Finality of the Dedekind reals. Every archimedean ordered field admissible for Omega embeds into RD.. It produces the two embeddings at source lines 659-660.

Theorem 11.2.14. *Every archimedean ordered field which is admissible for* Ω *is a subfield of* $\mathbb{R}_{\mathsf{d}}$.

Lemma 11.2.15. *If* F *is admissible for* Ω *then so is its Dedekind completion.*

Corollary 11.2.16. *The Dedekind reals are Dedekind complete: for every real-valued Dedekind cut* (L, U) *there is a unique* $x : \mathbb{R}_{\mathsf{d}}$ *such that* $L(y) = (y < x)$ *and* $U(y) = (x < y)$ *for all* $y : \mathbb{R}_{\mathsf{d}}$.

Proof. By Lemma 11.2.15 the Dedekind completion $\bar{\mathbb{R}}_{\mathtt{d}}$ of $\mathbb{R}_{\mathtt{d}}$ is admissible for Ω, so by Theorem 11.2.14 we have an embedding $\bar{\mathbb{R}}_{\mathtt{d}} \to \mathbb{R}_{\mathtt{d}}$, as well as an embedding $\mathbb{R}_{\mathtt{d}} \to \bar{\mathbb{R}}_{\mathtt{d}}$. But these embeddings must be isomorphisms, because their compositions are order-preserving field homomorphisms which fix the dense subfield $\mathbb{Q}$, which means that they are the identity. The corollary now follows immediately from the fact that $\bar{\mathbb{R}}_{\mathtt{d}} \to \mathbb{R}_{\mathtt{d}}$ is an isomorphism. $\square$